Problem Selection Analysis

Energy & Environment Case Study

This document details the evaluation and scope assessment for the Energy & Environment case track. It outlines the selected primary problem statement, justification for its implementation, and formal rationale for rejecting alternative proposals.

Executive Summary

Category	Problem Title	Decision Status	Primary Rationale
Data Integration	The Missing Meter	SELECTED	Essential baseline requirement; highly scalable with measurable proof of concept.
Load Classification	The Safety Boundary	REJECTED	High domain-specific electrical risk; violates safety boundary constraints.
Behavioral Incentives	User Nudges	REJECTED	Unpredictable psychological factors; proven history of short-lived impact.
Solar Optimization	Pre-Cooling	REJECTED	Requires unsupported HVAC assumptions; breaches direct control boundaries.

Selected Problem Statement

Data Integration: The Missing Meter

Problem Statement:
How can the campus accurately track and predict building-level energy consumption when some buildings provide 15-minute interval data while others only provide monthly totals?

Strategic Justification

- **Foundation Before Optimization:** Because some campus buildings report data in 15-minute intervals while others only provide monthly totals, any attempt to optimize the grid is essentially guessing. Fixing the data visibility gap is the mandatory first step before any control systems can be safely implemented.
- **Real-World Scalability:** Uneven data quality is a universal challenge in infrastructure and governance, not just an isolated campus issue. By building a system that infers missing information from partial data, we are solving a highly scalable, recognizable problem.
- **Clear, Testable Scope:** The hackathon requires a meaningful, working slice that can be tested. This problem provides a clear baseline for success: we can quantitatively test how accurately our solution predicts missing data without relying on unpredictable human behavior or requiring physical hardware.

Rejected Problem Statements

1. Load Classification: The Safety Boundary

Problem Statement:
How can facilities staff safely automate load-shifting during the evening demand spike when the system cannot currently distinguish between flexible cooling loads and critical, uninterruptible lab equipment?

Rationale for Rejection

- **High Domain-Specific Risk:** Accurately distinguishing between a flexible cooling load and critical, uninterruptible lab equipment requires deep domain expertise in electrical signatures. This risks bogging the project down in electrical engineering rather than general problem-solving.
- **Strict Safety Constraints:** Misclassifying a critical load and attempting to automate it would violate the core safety requirement of not disrupting research or student welfare.

2. Behavioral Incentives: User Nudges

Problem Statement:
How can the campus sustain energy reductions during peak evening hours when past awareness campaigns only produced short-lived behavioral changes among students and hostel residents?

Rationale for Rejection

- **Unpredictable Human Variables:** Relying on user apps and behavioral nudges introduces human psychology into the loop, making it very difficult to mathematically prove the solution's effectiveness during a short hackathon.
- **Historical Precedent of Failure:** The case explicitly notes that past awareness campaigns only produced a short-lived reduction in consumption. Defending a new behavioral approach as a permanent, sustainable fix would be an uphill battle.

3. Solar Optimization: Pre-Cooling

Problem Statement:
How can the campus reduce grid reliance during the 4 PM consumption spike when rooftop solar output drops precisely as the demand begins, and there is no system to optimally pre-cool buildings?

Rationale for Rejection

- **Excessive Physical Assumptions:** Designing a system to pre-cool buildings requires making massive assumptions about the campus's HVAC infrastructure, thermal mass, and insulation, which are outside the provided evidence.
- **Boundary Violations:** The case explicitly warns against assuming direct control over electrical infrastructure. Building an automated pre-cooling system forces the prototype to remain purely theoretical to avoid crossing safety and authorization boundaries.